From Counterexamples to Inductive Proofs

A Tutorial on SBV

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SBV is a Haskell library for reasoning about programs with SMT solvers. This tutorial is a guided path through it for Haskell programmers with no prior background in automated reasoning. We begin with one-line theorems and counterexamples, write specifications that are independent of the code they check, search for satisfying or optimal values, and program a live solver session from Haskell. We then define symbolic algebraic datatypes and recursive functions, and use induction and equational reasoning, expressed as ordinary Haskell code, to establish properties of recursive programs. We show how proof scripts can be reused at different element types and parameterized by operations. A detailed proof of Kadane's maximum-segment-sum algorithm shows how to find the right invariant, and an overview of a verified tautology checker shows how such proofs scale. Finally, we compile a verified function to C. Along the way we explain the boundary between Haskell evaluation and symbolic reasoning, which often determines how a problem must be expressed. Executable listings are extracted from checked Haskell sources; a companion validates solver results, runs the proofs and exercise solutions, and exhaustively tests the generated byte adder.

Additional Key Words and Phrases: Haskell, SMT, symbolic programming, theorem proving, induction, tutorial

1 INTRODUCTION

Haskell programmers already state properties of their code: QuickCheck properties [Claessen and Hughes 2000] can begin as ordinary functions that return `Bool`, and a test campaign searches for inputs that falsify them. SBV [Erkök 2026] lets the same programmer ask a stronger question. A property over *symbolic* values is translated to logic and handed to a solver for satisfiability modulo theories (SMT). Such a solver combines logical reasoning with theories such as integer arithmetic and fixed-width bit-vectors. When it can settle the query, it proves the property for every input or produces a concrete counterexample:

```
ghci> prove $ \x -> abs (x :: SInteger) .>= 0
Q.E.D.
ghci> prove $ \x -> abs (x :: SInt8) .>= 0
Falsifiable. Counter-example:
  s0 = -128 :: Int8
```

The two properties differ only in a type. Over mathematical integers, absolute values are non-negative. Over signed bytes, 128 is not representable, so `abs (-128)` is −128. The solver did not test this value by luck: it solved a constraint whose domain contains all 256 inputs and reported a value that fails. This does not require enumerating those inputs.

This tutorial is for Haskell programmers who want to use solvers in that way and then go further. It assumes familiarity with type classes and monads but no knowledge of SMT, logic, or theorem proving. Each section introduces one idea and builds on the previous ones:

- Sections 2–5 cover proofs, counterexamples, specifications, constraint solving, and optimization with multiple objectives.
- Section 6 explains the boundary between Haskell evaluation and symbolic reasoning, which governs how recursion and data structures behave.
- Section 7 models memory with symbolic arrays and shows why aliasing matters when reasoning about updates.
- Sections 8–9 program a live solver session and introduce symbolic algebraic datatypes.

- Sections 10–13 prove properties of recursive functions with SBV's theorem-proving layer, TP, including polymorphic and higher-order proofs, and ending with a detailed proof of Kadane's algorithm and an overview of a verified tautology checker.
- Section 14 compiles a verified function to C, and Section 15 explains enough of the implementation to make SBV's behavior predictable.

Section 16 collects the pitfalls in one place, and Section 17 relates SBV to other tools.

The aim is to teach a reusable method: choose a semantic domain, write a specification, inspect counterexamples, and introduce proof structure when automatic reasoning is insufficient. We verify functions expressed with SBV's symbolic operations; SBV does not translate arbitrary existing Haskell source into logic. In particular, a proof about mathematical integers or finite lists must not be silently transferred to machine integers, partial computations, or infinite lists.

Following along. The companion was checked with the SBV 14.10 source tree and GHC 9.14.1. The solvers were Z3 5.3.2 and CVC5 1.4.2; the execution log records the source revision and exact tool versions. Z3 [de Moura and Bjørner 2008] is the default solver. The tautology-checker development also uses CVC5 [Barbosa et al. 2022]. Install the solvers on the `PATH`; installing the Haskell package alone does not install them. Solver support varies by theory, so choosing a different solver may require changing the model.

After building SBV as described in the companion's README, start GHCi from the repository root:

```
cabal exec --offline -- ghci -ignore-dot-ghci -i -package sbv \
  Artifacts/Tutorial/examples/TutorialExamples.hs
```

This loads all the examples and their imports; `-i` ensures that GHCi uses the built SBV package. To write a separate module, use the following extensions and imports (some are only needed in later sections):

```
{-# LANGUAGE DataKinds           #-}
{-# LANGUAGE FlexibleInstances   #-}
{-# LANGUAGE OverloadedLists     #-}
{-# LANGUAGE QuasiQuotes         #-}
{-# LANGUAGE ScopedTypeVariables #-}
{-# LANGUAGE TemplateHaskell     #-}
{-# LANGUAGE TypeApplications    #-}
```

```
import Data.SBV
import Data.SBV.Control
import qualified Data.SBV.List as SL
import Data.SBV.TP
import Data.SBV.Tools.CodeGen
```

`OverloadedLists` lets literals such as `[1,2,3]` denote symbolic lists; `QuasiQuotes` enables `sCase`; `TemplateHaskell` enables datatype generation. `DataKinds` and `TypeApplications` support named quantifiers and explicit instantiation in proofs. `ScopedTypeVariables` permits the type annotations on lambda-bound variables used below. In GHCi, enable extensions for interactive expressions with:

```
:set -XOverloadedLists -XScopedTypeVariables
:set -XDataKinds -XTypeApplications
```

The companion extracts executable listings from its own source and the SBV documentation modules. Transcripts and explicitly abbreviated listings are presented separately. Model values and their order can vary between solver versions; the validation checks their meaning rather than their printed form.

2 FIRST PROOFS AND COUNTEREXAMPLES

2.1 Symbolic values and their types

SBV's central type is `SBV a`, a symbolic value whose concrete counterpart has type `a`. Abbreviations name the common cases: `SBool`, `SInteger` (unbounded integers), `SWord8` through `SWord64` and `SInt8` through `SInt64` (bit-vectors with wrap-around arithmetic), `SReal` (exact real numbers), `SFloat` and `SDouble` (IEEE-754), `SString`, and `SList a`. There are also arrays, sets, tuples, `Maybe`, `Either`, and user-defined algebraic datatypes.

These types are not interchangeable encodings of one numeric type; each is a different mathematical problem. The introduction showed that the choice between `SInteger` and `SInt8` changes a theorem into a false statement. Floating point has its own surprises:

```
ghci> prove $ \(x :: SFloat) -> x .== x
Falsifiable. Counter-example:
  s0 = NaN :: Float
```

Equality in IEEE-754 is not reflexive, and the solver knows it. Choosing a symbolic type is choosing the semantics the solver will use, so it deserves the same care as choosing a type for a production program.

2.2 Operators with a dot

Arithmetic on symbolic values uses the ordinary `Num` class, so `x + 1` and `2 * y` work as expected. Comparisons cannot reuse `Eq` and `Ord`: their methods return a Haskell `Bool`, which must be known when Haskell evaluates it, while the result of comparing symbolic values may be unknown. SBV therefore provides `.==`, `./=`, `.<`, `.<=`, `.>`, and `.>=`, which return `SBool`. Boolean connectives are `.&&`, `.||`, `sNot`, and implication `.=>`.

For the same reason, Haskell's `if-then-else` cannot branch on an `SBool`. The function `ite :: Mergeable a => SBool -> a -> a -> a` builds a symbolic choice that the solver resolves. This distinction between *Haskell control* and *symbolic control* recurs throughout the tutorial. When all arguments are known, symbolic operations simply compute: `ite ((3 :: SInteger) .> 2) 10 20 :: SInteger` evaluates to the literal `10` without involving a solver.

2.3 What prove does

Calling `prove` on a function of symbolic arguments creates a fresh unknown for each argument, evaluates the function to obtain a symbolic formula $P(\vec{x})$, and asks the solver whether $\neg P(\vec{x})$ is satisfiable. If it is not, P holds for every input and SBV prints `Q.E.D.` If it is, the satisfying assignment is a counterexample. A third outcome is possible: for undecidable or very hard problems, the solver may give up or time out. SBV reports this as an unknown result, which is neither a proof nor a refutation.

Unknowns can also be created by name inside the `Symbolic` monad, which gives readable counterexamples and allows assumptions:

Table 1. Entry points into SBV. All of them consume the same symbolic programs.

Interface	You supply	Successful result	Section
`prove`	A property	Validity or a counterexample	2–3
`sat`, `allSat`	Constraints on unknowns	One or all satisfying models	4
`optimize`	Constraints and objectives	Optimal models or objective bounds	5
`query`	A Haskell program driving the solver	Whatever the program computes	8
TP	Lemmas, induction, calculations	Checked proofs with dependencies	10–13
`compileToC`	A symbolic function	C source code	14

Solver calls can also be inconclusive. Enumeration can fail to terminate, and an optimization objective need not have an attained optimum.

```
prove $ do x <- sInteger "x"
           y <- sInteger "y"
           constrain $ x .> 0 .&& y .> 0
           pure $ x * y .> 0
```

Here `constrain` restricts attention to inputs that satisfy the assumption. A contradictory assumption makes every property provable, so assumptions should be kept as weak as the property permits. For assumptions A and conclusion P, `prove` searches for a model of $A \wedge \neg P$; it does not establish that A is feasible. For example,

```
ghci> prove $ \(x :: SWord8) -> x .> 255 .=> x .== 0
Q.E.D.
ghci> sat $ \(x :: SWord8) -> x .> 255
Unsatisfiable
```

The first answer is a vacuous proof: no byte satisfies its premise. Checking the premise with `sat` exposes the modeling mistake. Here and below, fresh arguments have the default universal interpretation under `prove` and existential interpretation under `sat`. Explicit mixed quantifiers are possible but outside this tutorial.

Table 1 lists the main ways of using a symbolic model. The rest of the tutorial visits each of them.

3 WRITING SPECIFICATIONS

A proof is only as good as the property it proves. The most useful properties compare an implementation with a specification written in a different, simpler way. Consider addition on unsigned bytes that saturates at 255 instead of wrapping. An implementation can avoid overflow by testing before it adds, while the specification can use unbounded integers, where overflow cannot happen:

```
satAdd :: SWord8 -> SWord8 -> SWord8
satAdd x y = ite (x .> 255 - y) 255 (x + y)

satAddSpec :: SWord8 -> SWord8 -> SBool
satAddSpec x y = sFromIntegral (satAdd x y) .== ite (total .> 255) 255 total
  where total = sFromIntegral x + sFromIntegral y :: SInteger
```

The guard `x .> 255 - y` cannot overflow, because `255 - y` is representable for every byte `y`, and on the other branch `x + y` does not wrap. In the specification, `sFromIntegral` converts each byte to an integer *before* adding. Converting the already wrapped byte sum would repeat the implementation's arithmetic and could hide its mistakes.

```
ghci> prove satAddSpec
Q.E.D.
```

The proof covers all 65,536 input pairs. A tempting alternative implementation shows why the independent specification matters:

```
wrongAdd :: SWord8 -> SWord8 -> SWord8
wrongAdd x y = ite (x + y .> 255) 255 (x + y)

wrongAddSpec :: SWord8 -> SWord8 -> SBool
wrongAddSpec x y = sFromIntegral (wrongAdd x y) .== ite (total .> 255) 255 total
  where total = sFromIntegral x + sFromIntegral y :: SInteger
```

```
ghci> prove $ do { x <- sWord8 "x"; y <- sWord8 "y"; pure (wrongAddSpec x y) }
Falsifiable. Counter-example:
  x = 223 :: Word8
  y = 124 :: Word8
```

The sum $223 + 124 = 347$ wraps to 91, and no byte is greater than 255, so the saturation branch is never taken. Had the specification also used byte arithmetic, it would have agreed with the bug. Your solver may report a different pair; any pair whose sum exceeds 255 is a valid counterexample, and the companion checks the reported pair with ordinary Haskell arithmetic.

Exercise 1. Write `satSub :: SWord8 -> SWord8 -> SWord8`, subtraction that saturates at 0, together with a specification over `SInteger`, and prove that they agree.

4 FINDING VALUES

Proving a property asks whether every input satisfies it. The dual question asks whether *some* input satisfies a constraint, and which one. The function `sat` returns one satisfying model, and `allSat` enumerates all of them. Which bytes are unchanged by squaring?

```
idempotentBytes :: IO AllSatResult
idempotentBytes = allSat $ \(x :: SWord8) -> x * x .== x
```

```
ghci> idempotentBytes
Solution #1:
  s0 = 1 :: Word8
Solution #2:
  s0 = 0 :: Word8
Found 2 different solutions.
```

The answer is a small theorem in disguise: $x(x-1) \equiv 0 \pmod{256}$, and since x and $x-1$ are coprime and $256 = 2^8$, one of them must be divisible by 256. `allSat` repeatedly finds a model and excludes it before searching again. The finite byte domain makes a complete enumeration possible here; on an unbounded domain, asking for all models may never finish.

5 OPTIMIZATION

The function `optimize` ranks feasible solutions by named objective *measures*: `minimize` a cost or `maximize` a profit. SBV's `Metric` class supports symbolic integers, reals, and bit-vectors, among others. These objectives differ from the decreasing measures used later to establish termination. A workshop makes two products with profits 2 and 3, subject to two resource limits:

```
production :: IO OptimizeResult
production = optimize Lexicographic $ do
  x <- sInteger "x"
  y <- sInteger "y"
  constrain $ x .>= 0 .&& y .>= 0
  constrain $ x + 2 * y .<= 10
  constrain $ 3 * x + y .<= 15
  maximize "profit" (2 * x + 3 * y)
```

Running `production` gives $x = 4$, $y = 3$, and profit 17. Here `constrain` sets hard requirements. Over `SInteger` this is integer linear programming; using `SReal` instead allows fractional production quantities. The resource bounds ensure that the maximum is attained. In general, check for infeasibility, unbounded objectives, and unattained bounds: maximizing a real x subject to $x < 1$ has supremum 1 but no optimal model.

Combining measures. The first argument of `optimize` selects an `OptimizeStyle`:

- `Lexicographic` optimizes objectives in declaration order. Once the first objective is optimal, the second breaks ties, and so on. For example, maximize profit first, then minimize the number of units produced without sacrificing profit.
- `Independent` uses Z3's *box* mode: optimize each objective separately under the same constraints, returning a model for each. These individual optima need not be jointly attainable.
- `Pareto Nothing` enumerates nondominated objective vectors: improving one objective requires worsening another. Their set is the *Pareto front*. Use `Pareto (Just n)` to request at most n such points; an infinite front makes unrestricted enumeration nonterminating.

For comparison, maximize nonnegative integers x and y with $x + y \leq 10$. Lexicographic order x then y gives $(10, 0)$; box gives separate maxima $x = 10$ and $y = 10$; the Pareto front consists of the eleven pairs with $x + y = 10$. Box does not promise the infeasible pair $(10, 10)$.

Bounds in an extension field. Extension-field results apply to both `SInteger` and `SReal`: unbounded objectives can yield `oo` or `-oo`. Real objectives can additionally require infinitesimals. For example, independently minimize and maximize a positive real:

```
realBounds :: IO OptimizeResult
realBounds = optimize Independent $ do
  x <- sReal "x"
  constrain $ x .> 0
  minimize "lower" x
  maximize "upper" x
```

Both independent results report these extension-field bounds in the tested configuration:

```
lower = epsilon :: Real
upper =      oo :: Real
```

Here `epsilon` denotes a positive infinitesimal: the infimum is 0 but is unattained. It is not a small floating-point tolerance. The symbol `oo` denotes positive infinity: the maximum is unbounded. Neither is a concrete real assignment to `x`. SBV exposes such bounds through `getModelObjectives` as `GeneralizedCV` values, using `ExtendedCV` for extension-field expressions. Changing `sReal` to `sInteger` in this example makes the minimum 1, while the maximum remains `oo`.

Strict-bound pitfall. This output is version-specific. Z3 does not guarantee correct infinitesimal results for optimization with strict inequalities such as `x .> 0`, especially when theories are mixed. Issue 5314 reports this very minimization returning 1 instead of an infinitesimal, without an unknown result [Erkök and Bjørner 2021]. Check open-bound behavior for the solver and formulation in use; a feasible model or a successful solver status alone does not establish optimality. Replacing $>$ by $\geq$ changes the problem.

Solver support and arithmetic limits. Z3 is currently the only optimization backend supported by SBV. SMT-LIB has not standardized optimization commands; SBV uses Z3's extensions for objectives and priorities [Microsoft 2026b]. Optimization is therefore less portable across solvers than ordinary satisfiability checking.

Nonlinear arithmetic is difficult in both optimization and SMT solving generally. Multiplying unknowns, as in `x * y`, leaves linear arithmetic; `3 * y` remains linear. Nonlinear integer satisfiability is undecidable, and Z3 may return unknown or fail to finish [Microsoft 2026a]. Optimization must also establish that no better solution exists. Z3 has specialized polynomial real procedures, including improved optimization support since 5.3, but this gives no general guarantee for nonlinear integer, mixed, or quantified problems [Microsoft 2026b]. Prefer linear formulations where faithful to the problem, set resource limits, and distinguish a feasible candidate from an established optimum.

6 HASKELL STRUCTURE AND SYMBOLIC STRUCTURE

An SBV interaction separates two kinds of work: Haskell evaluates the program to build a symbolic formula, and the solver reasons about that formula. Query programs can alternate between these activities. Knowing which side handles which part of a program explains how recursion behaves.

6.1 Haskell lists of symbolic values

A Haskell list `[SInteger]` has a spine that Haskell owns and elements that may be symbolic. Functions such as `sum`, `map`, and `foldr` work on it unchanged, because they only rearrange the spine and apply `Num` operations to the elements:

```
nonNegSum :: Int -> Predicate
nonNegSum n = do
  xs <- mapM (\i -> sInteger ("x" ++ show i)) [1 .. n]
  pure $ sAll (.>= 0) xs .=> sum xs .>= 0
```

For example, `prove (nonNegSum 5)` establishes the property for five symbolic integers. Haskell unfolds `sum` into `x1 + x2 + x3 + x4 + x5` before the solver sees anything; the recursion has disappeared. This style is ideal for fixed-size problems: circuits, unrolled loops, puzzles on fixed boards. The price is that the theorem is about lists of length five, not all lists.

6.2 Symbolic lists and recursive functions

The type `SList Integer` denotes a *finite* sequence whose length and contents may both be unknown. There is no fixed length bound, but infinite lists and partial values are outside this domain.

Throughout the tutorial, a theorem about "every list" means every finite list of values in the modeled element type (Section 15.2). To state such a property, we need a function whose recursion is part of the logic, not something Haskell unfolds. Writing a naive recursive Haskell function over `SList` does not work: whether the list is empty is unknown, so Haskell would unfold both branches forever. SBV's `smtFunction` solves this problem. It gives the function a name, and recursive calls become calls to that named function in the logic:

```
sumL :: SList Integer -> SInteger
sumL = smtFunction "sumL" $ \xs -> [sCase| xs of
          []     -> 0
          y : ys -> y + sumL ys
       |]
```

The `sCase` quasi-quoter provides pattern matching on symbolic values. The definition reads like ordinary Haskell, but SBV translates it to a single recursive SMT-LIB definition [Barrett et al. 2025]. Because `sumL` is a function of symbolic values, it still computes on literals: `sumL [1,2,3]` evaluates to 6 without a solver. When a recursive definition is registered for symbolic reasoning, SBV checks termination; for `sumL`, it automatically finds the list length as a decreasing, nonnegative measure. Termination makes the definition admissible; it does not prove arbitrary properties of it.

Now try the obvious generalization of `nonNegSum`:

```
sumWithoutInduction :: IO ThmResult
sumWithoutInduction =
  proveWith (z3 { solverSetOptions = [SetTimeOut 5000] }) $
    \xs -> SL.all (.>= 0) xs .=> sumL xs .>= 0
```

The property is true, but ordinary SMT reasoning about recursive definitions does not reliably establish it. The timeout above limits each solver call to five seconds; an unknown answer is expected with the tested configuration. It says nothing about the truth of the property. Section 10 supplies an induction argument that works for lists of every finite length.

The general rule is: *recursion over Haskell structure happens while building the problem; recursion through* `smtFunction` *becomes part of the problem.* The former fixes the amount of unfolding before solving; the latter allows statements about structures of unbounded size. Neither representation guarantees that the resulting solver query will be easy.

7 ARRAYS AND SYMBOLIC MEMORY

An `SArray Integer Integer` is a total mapping from integer addresses to integer values. Use `sArray "mem"` to create an unknown array and `readArray mem i` to read a symbolic address. Updates are functional: `writeArray mem i v` returns an updated array, preserving `mem` as the previous state.

The array theory supplies read-after-write laws: the updated address returns the new value; all other addresses retain their values. Equality is *extensional*: equal arrays agree at every address, and conversely [Tinelli 2024]. Thus writes to distinct addresses commute:

```
writesCommute :: SArray Integer Integer
              -> SInteger -> SInteger -> SInteger -> SInteger -> SBool
writesCommute a i j x y =
  i ./= j .=>
    writeArray (writeArray a i x) j y .==
    writeArray (writeArray a j y) i x
```

```
ghci> prove writesCommute
Q.E.D.
```

The proof covers every initial array, both addresses, and both values. The solver establishes equality of entire arrays directly, without traversal or induction.

Omit `i ./= j` and the property is false: the addresses may *alias*. For example, with $i = j = 0$, $x = 1$, and $y = 2$, the left side stores 2 at address 0 while the right side stores 1. The last write wins, regardless of the initial contents. The companion obtains a counterexample from the solver and checks its updates independently using Haskell association-list lookup.

Bounds and the modeling domain. An SMT array has no built-in length, allocation status, or out-of-bounds error. To model a buffer of length `n`, require `0 .<= i .&& i .< n` for a valid access; to verify safety, prove that the program establishes this condition before accessing the buffer. Merely assuming it would exclude invalid accesses from the proof. `constArray v` initializes every address to `v`; a free array has arbitrary but consistent contents.

An integer-indexed array has an infinite index domain. This does not contradict the finite-list and recursive-datatype restriction discussed in Section 15.2: an array is a total mapping, not a recursively constructed sequence. The domain follows the index type: `SArray Word8 Integer` has 256 addresses, whereas `SArray Integer Integer` has infinitely many. Neither represents a partial or infinite Haskell list.

Bulk updates. The function `lambdaArray` describes an array by its value at each index. The library example `Documentation.SBV.Examples.Misc.LambdaArray` uses it to model range initialization: return the new value within the chosen interval and `readArray mem i` outside it. This expresses both the changed region and preservation of the remaining memory without enumerating addresses.

8 PROGRAMMING THE SOLVER

The functions `prove`, `sat`, and `optimize` manage solver interactions on the caller's behalf. Many applications need a conversation: ask, inspect the answer, and ask a refined question. SBV's `query` mode hands control of a live solver session to a Haskell program. The following function finds the smallest integer greater than 10 that leaves remainder 3 when divided by 7 and remainder 4 when divided by 5:

```
smallest :: IO Integer
smallest = runSMT $ do
  x <- sInteger "x"
  constrain $ x .> 10
  constrain $ x `sMod` 7 .== 3
  constrain $ x `sMod` 5 .== 4
  query $ let descend best = do
                r <- checkSat
                case r of
                  Sat   -> do v <- getValue x
                              constrain $ x .< literal v
                              descend (Just v)
                  Unsat -> maybe (io $ fail "no solution") pure best
                  _     -> io $ fail "inconclusive"
          in descend Nothing
```

```
ghci> smallest
24
```

The constraints before `query` set up the problem as usual. Inside the query, `checkSat` asks for a model, and `getValue` reads the value of `x` in that model. That value is an ordinary `Integer`, so Haskell can branch on it and use it to build the next constraint, `x .< literal v`, which asks for a smaller solution. When no smaller solution exists, the last model found is the minimum. The `case` on the solver's answer is Haskell control; the constraint that it adds is symbolic. Keeping the two apart is what makes the solver usable as a component of a larger algorithm.

Three properties of this loop matter. First, successful candidates are strictly decreasing *integers* bounded below by 11. After the first candidate v, there can be at most $v-11$ further decreases. Thus the loop terminates provided each solver call returns. The existence of a minimum alone would not justify this reasoning over reals: the candidates could decrease forever without reaching it. Second, the solver may return the candidates in any order. Our loop does not depend on that order; it only relies on unsatisfiability at the end. Third, the loop handles an unknown answer separately: an inconclusive solver response is not evidence that no smaller value exists.

Queries can do much more: `push` and `pop` add and retract temporary assumptions, named constraints allow unsatisfiable cores to be extracted, and new unknowns can be created in the middle of a session with `freshVar`. The last point is less trivial than it looks: a new variable may have a type, such as a user-defined datatype, that the solver has not seen yet. SBV tracks the declarations each new term needs and sends them before the term is used, so queries compose without regard to the order in which types are first used.

Exercise 2. Change `smallest` to find the *largest* solution below 100. (The answer is 94.)

9 SYMBOLIC DATATYPES

User-defined algebraic datatypes become symbolic with one line of Template Haskell. Here is a tiny expression language and its evaluator:

```
data Expr = Num Integer
          | Add Expr Expr
          | Mul Expr Expr
          deriving Show
mkSymbolic [''Expr]

eval :: SExpr -> SInteger
eval = smtFunction "eval" $ \e -> [sCase| e of
          Num n   -> n
          Add a b -> eval a + eval b
          Mul a b -> eval a * eval b
       |]

puzzle :: IO SatResult
puzzle = sat $ \e -> eval e .== 42 .&& isMul e .&& sNot (isNum (getMul_1 e))
```

The call `mkSymbolic [''Expr]` generates the type `SExpr`, symbolic constructors (`sNum`, `sAdd`, `sMul`), recognizers (`isNum`, `isAdd`, `isMul`), and field accessors such as `getMul_1`. The evaluator is an `smtFunction`, so it is a recursive definition in the logic, and `sCase` matches on an expression whose constructor may be unknown. The puzzle asks the solver to *synthesize* an expression: a multiplication whose left argument is not a number, evaluating to 42.

```
ghci> puzzle
Satisfiable. Model:
  s0 = Mul (Mul (Num 1) (Num (-1))) (Num (-42)) :: Expr
```

The model is an ordinary Haskell value of type `Expr`, which we can inspect, print, or evaluate with a separate Haskell interpreter. The companion does exactly that to check the answer independently of the solver. Different solver versions may find different expressions.

The constructor test `isMul e` matters: an accessor such as `getMul_1 e` describes the intended field only when `e` has that constructor. Also, this is synthesis from a structural constraint and one evaluation constraint, not a proof that the expression implements a function on every input. The search has no explicit depth bound and is not guaranteed to terminate for arbitrary constraints.

The same datatype can be the subject of proofs: a property of `eval` over all expressions concerns finite expression trees of unbounded size, and may need induction. Infinite or partial trees are outside the model (Section 15.2). Induction is the topic of the next section.

10 PROOFS BY INDUCTION

SMT solvers decide many theories well, but an induction argument may be needed to connect recursive definitions to a theorem. SBV's TP layer (in `Data.SBV.TP`) lets the programmer write the argument in Haskell and has the solver check each step. The programmer decides *what* to prove and *how* to break it into steps; the solver decides whether each step is valid.

10.1 A first inductive proof

We return to the property that eluded `prove` in Section 6:

```
sumNonNeg :: TP (Proof (Forall "xs" [Integer] -> SBool))
sumNonNeg = induct "sumNonNeg"
  (\(Forall xs) -> SL.all (.>= 0) xs .=> sumL xs .>= 0) $
  \ih (x, xs) -> [SL.all (.>= 0) (x .: xs)]
              |- sumL (x .: xs) .>= 0
              =: x + sumL xs .>= 0
              ?? ih
              =: sTrue
              =: qed
```

The type `Proof (Forall "xs" [Integer] -> SBool)` records the *quantifier shape*: a property of every finite integer list `xs`. The actual predicate is the first function argument to `induct`, stored in the proof value; it is not encoded in this Haskell type. Two different properties can therefore have the same `Proof` type. The name in `Forall` identifies the quantified variable for output and instantiation. `TP` is the monad in which proof scripts are built, and `runTP` executes a script and returns its proof object. Typechecking the script alone does not check the theorem.

The function `induct` applies list induction: SBV checks the base case (the empty list) automatically and gives the user the inductive step. In that step, `ih` is the induction hypothesis, a proof of the property for `xs`, and the goal is to show it for `x .: xs`. These cases cover lists built by finitely many cons operations from the empty list. They do not establish the property for infinite Haskell lists. Section 15.2 discusses infinite lists and explains why this restriction also applies to recursive algebraic datatypes.

The step is a *calculation*. Its first line lists the assumptions, here the precondition for the longer list, and `|-` starts a chain of expressions connected by `=:`. Each link must be justified by the definitions, the assumptions, and any hints supplied with `??`. The chain ends with `qed`. A hint written

between two expressions justifies the transition *to the following expression*; `?? ih` is not a new assumption about the entire theorem. In this example the expressions being equated are Boolean formulas, and the calculation reduces the target formula to `sTrue`. Running the proof shows each checked step (the final proof-object display is omitted here):

```
ghci> runTP sumNonNeg
Inductive lemma: sumNonNeg
  Step: Base                 Q.E.D.
  Step: 1                    Q.E.D.
  Step: 2                    Q.E.D.
  Result:                    Q.E.D.
Functions proven terminating: sbv.foldr, sumL
```

Step 1 unfolds `sumL` once. Step 2 uses the hypothesis: `x` is nonnegative by assumption and so is `sumL xs`, so the sum is too. The `Result` line checks that the chain, together with the base case, establishes the theorem as stated. The last line reports that the recursive functions used, including the library's `foldr` that underlies `SL.all`, were checked for termination. Section 15.1 explains how TP enforces this requirement.

10.2 What the solver checks

A calculation $e_0 = e_1 = \cdots = e_n$ under assumptions A, with hints H_i attached to step i, generates one solver query per step:

$$A \wedge H_i \wedge \neg(e_i = e_{i+1}) \quad \text{must be unsatisfiable.}$$

The function definitions are available in these queries, and SBV separately checks that the calculation establishes the stated result under its precondition. Thus writing an unrelated true equality does not prove the theorem. Breaking a proof into local obligations often makes it easier to solve, although any individual obligation can still be hard. If a step fails, inspect whether the solver found a counterexample or returned unknown. Then check the assumptions and hint instantiations, or split the step with intermediate expressions.

10.3 Generalizing the statement

Reversing a list with an accumulator is a standard optimization:

```
revAcc :: SList Integer -> SList Integer -> SList Integer
revAcc = smtFunction "revAcc" $ \acc xs -> [sCase| xs of
           []     -> acc
           x : xt -> revAcc (x .: acc) xt
         |]
```

We expect `revAcc [] xs` to equal `SL.reverse xs`. The direct attempt is instructive:

```
revWeak :: TP (Proof (Forall "xs" [Integer] -> SBool))
revWeak = induct "revWeak"
  (\(Forall xs) -> revAcc [] xs .== SL.reverse xs) $
  \ih (x, xs) -> [] |- revAcc [] (x .: xs)
                    =: revAcc [x] xs
                    ?? ih
                    =: SL.reverse xs SL.++ [x]
                    =: SL.reverse (x .: xs)
                    =: qed
```

```
ghci> runTP revWeak
Inductive lemma: revWeak
  Step: Base                 Q.E.D.
  Step: 1                    Q.E.D.
  Step: 2
*** Failed to prove revWeak.2.
```

Step 2 tries to use the induction hypothesis, but the hypothesis is about `revAcc [] xs`, while the expression contains `revAcc [x] xs`. After one step of recursion, the accumulator is no longer empty, and the hypothesis says nothing about it. The displayed failure comes from a five-second timeout on each solver call in the companion. It is a failure to complete this proof attempt, not a counterexample to reversal correctness. An unrestricted solver search might continue indefinitely.

The fix is the oldest trick in inductive proofs: prove a more general statement. The theorem must describe `revAcc` for *every* accumulator, so that the hypothesis can be used at whatever accumulator the recursion produces:

```
revAccCorrect :: TP (Proof (Forall "xs" [Integer] -> Forall "acc" [Integer] -> SBool))
revAccCorrect = induct "revAccCorrect"
  (\(Forall xs) (Forall acc) -> revAcc acc xs .== SL.reverse xs SL.++ acc) $
  \ih (x, xs) acc -> [] |- revAcc acc (x .: xs)
                        =: revAcc (x .: acc) xs
                        ?? ih
                        =: SL.reverse xs SL.++ (x .: acc)
                        =: (SL.reverse xs SL.++ [x]) SL.++ acc
                        =: SL.reverse (x .: xs) SL.++ acc
                        =: qed
```

Here `acc` is universally quantified, and the induction hypothesis `ih` holds for all accumulators. When it is supplied as a hint, the solver instantiates it with `x .: acc`. The remaining steps use ordinary laws of `++` and `reverse`, using their sequence operations and definitions. The original statement is the special case `acc = []`:

```
reverseCorrect :: TP (Proof (Forall "xs" [Integer] -> SBool))
reverseCorrect = do
  generalized <- recall revAccCorrect
  lemma "reverseCorrect"
    (\(Forall xs) -> revAcc [] xs .== SL.reverse xs)
    [proofOf generalized]
```

`recall` obtains the generalized proof, reusing it if it is already cached in the current TP run. `proofOf` packages that proof as a dependency for `lemma`, which asks the solver to establish the specialization. This separates the reusable inductive result from the interface property callers need.

Exercise 3. Prove that `SL.length (revAcc acc xs) .== SL.length xs + SL.length acc`. Why must the accumulator remain universally quantified in the induction hypothesis, even when the intended caller starts with `[]`?

10.4 Polymorphic scripts, monomorphic proofs

Reversal preserves length regardless of the element type. We can express the proof once, using Haskell polymorphism:

```
reverseLength :: forall a. SymVal a => TP (Proof (Forall "xs" [a] -> SBool))
reverseLength = induct "reverseLength"
  (\(Forall xs) -> SL.length (SL.reverse xs) .== SL.length xs) $
  \ih (x, xs) -> [] |- SL.length (SL.reverse (x .: xs))
                    =: SL.length (SL.reverse xs SL.++ [x])
                    =: SL.length (SL.reverse xs) + 1
                    ?? ih
                    =: SL.length xs + 1
                    =: SL.length (x .: xs)
                    =: qed
```

The Haskell `forall a` makes the *script* polymorphic; `SymVal a` provides the symbolic representation of its elements. Before running it, we choose a monotype. For example, in GHCi:

```
runTP (reverseLength @Integer)
runTP (reverseLength @Word8)
```

Each invocation checks a separate theorem over that type's solver sort. The companion runs both.

This differs from `Forall "xs" [a]`, which quantifies over *values* inside the theorem after `a` has been chosen. A successful integer instance does not by itself certify every type instance of the script: SBV is not quantifying over Haskell types. Polymorphism lets us reuse the argument and its implementation, while each proof run establishes a monomorphic result. This distinction matters especially for arithmetic laws, whose truth may change between `Integer` and `Word8`.

11 HIGHER-ORDER PROOFS

Many useful list laws concern an operation supplied by the caller. The *bookkeeping law*, adapted here from `Documentation.SBV.Examples.TP.Lists`, says that reducing a concatenation equals reducing each chunk and then reducing those results:

$$reduce(concat\ xss) = reduce(map\ reduce\ xss), \qquad reduce = foldr\ f\ e.$$

Associativity of f and a two-sided identity e suffice. For integer addition with identity 0, this is "sum the elements" versus "sum the chunk sums." Commutativity is not assumed: the order of elements is preserved. Both the outer list `xss` and every chunk are finite; the law makes no claim about consuming an infinite stream of chunks.

11.1 Proving the bookkeeping law

We reuse a library lemma with the following import:

```
import qualified Documentation.SBV.Examples.TP.Lists as ListProofs
```

The proof has two dependencies: `foldrOverAppend`, and a helper `foldBase` proved in the companion. Their conclusions are, respectively,

$$foldr\ f\ e\ (xs\ ++\ ys) = foldr\ f\ (foldr\ f\ e\ ys)\ xs,$$
$$foldr\ f\ y\ xs = f\ (foldr\ f\ e\ xs)\ y.$$

The first needs no assumptions on f. The second requires associativity and the left-identity law. Its empty-list case uses $f\ e\ y = y$; its inductive step uses associativity to move y past the head. These lemmas expose exactly the two transformations the main induction needs:

```
bookKeeping :: forall a. SymVal a
            => SBV a -> (SBV a -> SBV a -> SBV a)
            -> TP (Proof (Forall "xss" [[a]] -> SBool))
bookKeeping e f = do
  let assoc = quantifiedBool $ \(Forall x) (Forall y) (Forall z) ->
                f x (f y z) .== f (f x y) z
      rUnit = quantifiedBool $ \(Forall x) -> f x e .== x
      lUnit = quantifiedBool $ \(Forall x) -> f e x .== x
      reduce = SL.foldr f e
  base <- foldBase e f
  over <- ListProofs.foldrOverAppend e f
  induct "bookKeeping"
    (\(Forall xss) -> assoc .&& rUnit .&& lUnit .=>
       reduce (SL.concat xss) .== reduce (SL.map reduce xss)) $
    \ih (xs, xss) -> [assoc, rUnit, lUnit] |- reduce (SL.concat (xs .: xss))
      =: reduce (xs SL.++ SL.concat xss)
      ?? over
      =: SL.foldr f (reduce (SL.concat xss)) xs
      ?? ih
      =: SL.foldr f (reduce (SL.map reduce xss)) xs
      ?? base `at` (Inst @"xs" xs, Inst @"y" (reduce (SL.map reduce xss)))
      =: f (reduce xs) (reduce (SL.map reduce xss))
      =: reduce (reduce xs .: SL.map reduce xss)
      =: reduce (SL.map reduce (xs .: xss))
      =: qed
```

`quantifiedBool` packages a quantified property as an `SBool`, so the algebraic laws can appear as assumptions in the calculation and as the premise of the theorem. They are not unconditionally added as user axioms. Induction is over the outer list of chunks: unfold `concat`, move the fold across append, use the induction hypothesis on the remaining chunks, then use `foldBase` to combine their result with that of the first chunk.

The left-identity assumption has a concrete purpose. Take $f\ x\ y = x$ and $e = 0$. This operation is associative and has right identity 0, but for `xss = [[],[1]]`, the two sides of the bookkeeping law are 1 and 0. The empty first chunk contributes an identity that must work on the left. The companion checks this counterexample as well as the proof.

11.2 An arbitrary operation at a fixed type

The parameters `e` and `f` are Haskell arguments to the proof script. To prove the law without choosing an implementation of the operation, give them fresh uninterpreted names:

```
bookKeepingAbstract :: IO (Proof (Forall "xss" [[Integer]] -> SBool))
bookKeepingAbstract = runTP $
  bookKeeping @Integer (uninterpret "unit") (uninterpret "combine")
```

With `uninterpret`, a logical constant or function has no defining equations. Here `combine` is a total function on integers; the proof must work for every interpretation satisfying the stated algebraic laws. The solver does not enumerate implementations of `combine`. SBV specializes its list operations to this logical operation and reasons about the resulting definitions. The higher-order Haskell interface thus supports a proof about an arbitrary operation at a fixed type. We do not pass a Haskell function as an ordinary `Forall` argument of the theorem.

Passing `0` and `(+)` instead gives a concrete instance. The companion also checks that instance and derives the unconditional sum-of-chunk-sums law. An uninterpreted operation represents a total mathematical function; it does not model divergence or exceptions in a Haskell callback.

11.3 Closures and higher-order functions

A callback may also depend on data from its surrounding scope. Ordinary Haskell allows `map (+k) xs`, but `SL.map (+k) xs` cannot capture a symbolic unknown `k` from the enclosing solver context in the callback's definition. Use an explicit `Closure` to pass that environment:

```
shiftEach :: SInteger -> SList Integer -> SList Integer
shiftEach k = SL.map Closure { closureEnv = k
                             , closureFun = \env x -> x + env
                             }
```

The environment `k` is symbolic data; `closureFun` is closed with respect to it because it receives `env` as an argument. SBV carries the environment through the generated recursive definition alongside the list. The companion proves that `shiftEach` preserves length for every offset and every finite integer list.

This restriction concerns captured symbolic unknowns, not literals or globally named logical symbols. For example, `SL.map (+1)` needs no explicit closure. The uninterpreted names above also need no captured environment: they belong to the logical signature. Higher-order arguments describe which operation to use; a `Closure` additionally supplies the changing symbolic data it needs.

12 CASE STUDY: KADANE'S ALGORITHM

The maximum segment sum problem asks for the largest sum of a contiguous segment of a list, where the empty segment has sum 0. For `[1, -2, 3, 4, -1, 2]`, the answer is 8, from `[3, 4, -1, 2]`. Kadane's algorithm [Bentley 1984] solves it in a single left-to-right pass. Its correctness is not obvious, which makes it a good example of finding and proving an invariant.

12.1 Specification

We specify the problem by its most direct recursive reading:

```
mss :: SList Integer -> SInteger
mss = smtFunction "mss" $ \xs -> [sCase| xs of
        []    -> 0
        _ : t -> mssBegin xs `smax` mss t
      |]

mssBegin :: SList Integer -> SInteger
mssBegin = smtFunction "mssBegin" $ \xs -> [sCase| xs of
             []    -> 0
             h : t -> 0 `smax` (h `smax` (h + mssBegin t))
           |]
```

`mssBegin xs` is the best sum of a segment that starts at the beginning of `xs`: either the empty segment (0), just the head, or the head followed by the best segment starting at the tail. `mss xs` is the best of `mssBegin xs` and the best segment of the tail: a segment either starts at the head or lies entirely in the tail. With unit-cost arithmetic and without memoizing calls across suffixes, these recurrences take quadratic time. Since they are symbolic functions, they also run on literal

lists, which is a convenient way to test a specification before reasoning about it: `mss [1, -2, 3, 4, -1, 2]` evaluates to 8.

To check that the recurrence matches the intended problem, we can independently enumerate segments with ordinary Haskell's `Data.List.inits` and `Data.List.tails`:

```
segmentOracle :: [Integer] -> Integer
segmentOracle xs = maximum [sum segment | suffix <- tails xs, segment <- inits suffix]
```

Every segment is a prefix of a suffix, and the empty segment ensures the maximum is defined even for `[]`. The companion compares both `mss` and `kadane` against this oracle on all 3,906 lists of length at most five over $\{-2, -1, 0, 1, 2\}$, plus the examples above. These tests check the connection to the intended specification; the theorem below establishes equivalence for every finite integer list.

12.2 Implementation

Kadane's algorithm maintains two accumulators while it scans the list: `meh`, the maximum sum of a segment ending at the current position, and `msf`, the maximum found so far:

```
kadane :: SList Integer -> SInteger
kadane xs = kadaneHelper xs 0 0

kadaneHelper :: SList Integer -> SInteger -> SInteger -> SInteger
kadaneHelper = smtFunction "kadaneHelper" $ \xs meh msf -> [sCase| xs of
                  []    -> msf
                  h : t -> let meh' = 0 `smax` (h + meh)
                               msf' = msf `smax` meh'
                           in kadaneHelper t meh' msf'
              |]
```

At each element `h`, the best segment ending here either extends the best segment ending at the previous position or is empty, so the new `meh` is `0 `smax` (h + meh)`. The best so far is updated if that value is larger. The allowance of an empty segment matters: an all-negative input returns 0. An API that requires a nonempty segment would need a different specification and initialization.

12.3 Finding the invariant

As with reversal, the theorem we want, `kadane xs .== mss xs`, concerns the initial accumulators `0` and `0`. An inductive proof will reach `kadaneHelper` with other accumulator values, so we must describe what `kadaneHelper xs meh msf` computes for *any* sensible `meh` and `msf`.

Think of the list as split into a prefix that has been processed and a suffix `xs` that has not. The best segment of the whole list lies in one of three places:

(1) entirely within the prefix: its sum is `msf`;
(2) entirely within the suffix: its sum is `mss xs`;
(3) across the boundary: it is a segment ending at the boundary, whose best sum is `meh`, followed by a segment starting at the boundary, whose best sum is `mssBegin xs`.

For example, split `[1,-2,3,4,-1,2]` after the 4. The processed prefix has $meh = 7$ and $msf = 7$, while the suffix has $mss = 2$ and $mssBegin = 1$. The three candidates are 7, 2, and $7 + 1 = 8$; the best segment crosses the boundary. So the invariant is

$$kadaneHelper(xs, meh, msf) = \max\big(msf,\ mss(xs),\ meh + mssBegin(xs)\big),$$

under the preconditions that make the accumulators meaningful: $meh \geq 0$ (the empty segment is always available) and $msf \geq meh$ (the best segment so far is at least as good as the best one ending here). Without the preconditions the statement is false: for an empty `xs`, the right side would include *meh* even when it exceeds *msf*.

12.4 The proof

Once the invariant is written down, the proof is short:

```
kadaneCorrect :: TP (Proof (Forall "xs" [Integer] -> SBool))
kadaneCorrect = do
  inv <- induct "kadaneInvariant"
           (\(Forall xs) (Forall meh) (Forall msf) ->
                    (meh .>= 0 .&& msf .>= meh)
                .=> kadaneHelper xs meh msf
                    .== msf `smax` mss xs `smax` (meh + mssBegin xs)) $
           \ih (a, as) meh msf ->
              [meh .>= 0, msf .>= meh]
           |- let meh' = 0 `smax` (a + meh)
                  msf' = msf `smax` meh'
              in kadaneHelper (a .: as) meh msf
              =: kadaneHelper as meh' msf'
              ?? ih `at` (Inst @"meh" meh', Inst @"msf" msf')
              =: msf' `smax` mss as `smax` (meh' + mssBegin as)
              =: qed

  calc "kadaneCorrect" (\(Forall xs) -> kadane xs .== mss xs) $
       \xs -> let z = 0 :: SInteger
              in [] |- kadane xs
                    =: kadaneHelper xs z z
                    ?? inv `at` (Inst @"xs" xs, Inst @"meh" z, Inst @"msf" z)
                    =: z `smax` mss xs `smax` (z + mssBegin xs)
                    =: mss xs `smax` mssBegin xs
                    =: mss xs
                    =: qed
```

The inductive step unfolds `kadaneHelper` once and applies the hypothesis at the new accumulators. Here the hint is more specific than in the earlier proofs: `ih `at` (Inst @"meh" meh', Inst @"msf" msf')` instantiates the quantified accumulators explicitly. The solver could search for the instantiation, but stating it makes the step easier and documents the argument. The remaining equality between the instantiated hypothesis and the invariant for `a .: as` is arithmetic over `smax`, which the solver checks without further help, using the definitions of `mss` and `mssBegin`. The preconditions are needed to instantiate the hypothesis: the new accumulators satisfy them because `meh'` is a maximum with 0 and `msf'` is a maximum with `meh'`.

The main theorem then follows by a calculation without induction. With both accumulators 0, the invariant gives $\max(0, mss(xs), mssBegin(xs))$, and the solver discharges the last two steps: `mss` is nonnegative and at least `mssBegin` by its definition.

```
ghci> runTP kadaneCorrect
Inductive lemma: kadaneInvariant
  Step: Base                          Q.E.D.
  Step: 1                             Q.E.D.
  Step: 2                             Q.E.D.
  Result:                             Q.E.D.
Lemma: kadaneCorrect
  Step: 1                             Q.E.D.
  Step: 2                             Q.E.D.
  Step: 3                             Q.E.D.
  Step: 4                             Q.E.D.
  Result:                             Q.E.D.
Functions proven terminating: kadaneHelper, mss, mssBegin
```

The proof establishes equality of the two functions. The linear traversal of `kadaneHelper` follows separately from its one recursive call on the tail per element; the theorem does not establish a running-time or space bound for Haskell evaluation, and integer arithmetic itself is not constant-cost in general.

12.5 Lessons

The effort in this proof went into one place: the invariant. Once we had explained *why* the algorithm works, as the three-way case analysis on where the best segment lies, the mechanical checking was immediate. This division of labor is typical of TP. The solver handles unfolding definitions, arithmetic, and case analysis on `smax`; the human supplies generalizations and intermediate statements. When a proof gets stuck, the useful question is rarely "which tactic is missing?" and usually "what is the statement that explains this step?"

13 CASE STUDY OVERVIEW: A VERIFIED TAUTOLOGY CHECKER

Larger developments are built from the same ingredients. SBV's documentation includes a verified tautology checker for propositional formulas in if-then-else form, after a classic development by Boyer and Moore [1979] that was later formalized in Imandra [Passmore nd]. We sketch its structure rather than its proofs; the full development is in the module `Documentation.SBV.Examples.TP.TautologyChecker`. The listings in this section are quoted from that module.

Formulas and normalization. A formula is a constant, a variable, or a conditional:

```
data Formula = FTrue
             | FFalse
             | Var { fVar   :: Integer }
             | If  { ifCond :: Formula
                   , ifThen :: Formula
                   , ifElse :: Formula
                   }
```

The checker first normalizes a formula so that no conditional appears in the condition position of another conditional, by repeatedly rewriting `If (If p q r) l r'` to `If p (If q l r') (If r l r')`. Abbreviated slightly, the definition is:

```
normalize :: SFormula -> SFormula
normalize = smtFunctionWithMeasure "normalize"
    ( \f -> tuple (ifComplexity f, ifDepth f)
    , [measureLemma ifDepthNonNeg, measureLemma ifComplexityPos, ...] )
  $ \f -> [sCase| f of
             If (If p q r) l r' -> normalize (sIf p (sIf q l r') (sIf r l r'))
             If c l r           -> sIf c (normalize l) (normalize r)
             _                  -> f
         |]
```

This function is the first new ingredient. Its termination is not structural: the rewrite in the first branch produces a *larger* formula. Instead of letting SBV guess a measure, the definition supplies one through `smtFunctionWithMeasure`: a pair ordered lexicographically. To see why this works, write C for `ifComplexity` and D for `ifDepth`. On constants and variables, $C = 1$ and $D = 0$; on conditionals,

$$C(\mathit{If}\ c\ l\ r) = C(c)\big(C(l) + C(r)\big), \qquad D(\mathit{If}\ c\ l\ r) = 1 + D(c).$$

The distributive rewrite preserves C exactly and reduces D by one. Recursing into either branch of an ordinary conditional strictly reduces C, because every complexity is positive. Thus each recursive call decreases the pair, even though tree size may increase. Each `measureLemma` supplies a proved fact needed for these checks, such as positivity of C. Termination and truth preservation remain separate obligations.

Checking. On a normalized formula, the checker walks down the conditionals, recording an assumed value for each variable the first time it is tested. When it meets a variable that was already assigned, it follows the corresponding branch; otherwise it checks both branches, once under each assumption. The function `isTautology f` normalizes `f` and runs this walk with no assumptions.

The theorems. Soundness says that a formula accepted by the checker evaluates to true under every assignment:

```
soundness :: TP (Proof (Forall "f" Formula -> Forall "bindings" [Binding] -> SBool))
soundness = do
  tie <- recallWith cvc5 tautologyImpliesEval
  nrt <- recall normalizeRespectsTruth
  nc  <- recall normalizeCorrect

  calc "soundness"
       (\(Forall f) (Forall bindings) -> isTautology f .=> eval f bindings) $
       \f bindings -> [isTautology f]
                   |- eval f bindings
                   ?? nrt `at` (Inst @"f" f, Inst @"bs" bindings)
                   =: eval (normalize f) bindings
                   ?? nc  `at` Inst @"f" f
                   ?? tie `at` (Inst @"f" (normalize f), Inst @"a" bindings, Inst @"b" [])
                   =: sTrue
                   =: qed
```

Completeness says that if the checker rejects a formula, a companion function `falsify` produces an assignment under which the formula is false:

```
completeness :: TP (Proof (Forall "f" Formula -> SBool))
completeness = do
  ch  <- recall completenessHelper
  nrt <- recallWith z3 normalizeRespectsTruth

  lemma "completeness"
        (\(Forall f) -> sNot (isTautology f) .=> sNot (eval f (scex (falsify f))))
        [proofOf ch, proofOf nrt]
```

Both top-level proofs are short because the work is in their lemmas. The development contains 21 TP lemmas. Several prove properties of variable lookup when assignments are extended; others establish that normalization produces a normal formula and preserves meaning. The central inductive lemmas, such as `tautologyImpliesEval`, use *strong induction* via `sInduct`: the hypothesis is available for every formula with a smaller measure, not only for immediate subformulas, which matches the shape of the recursion in `isTautology'`. The lemmas use `recall` to reuse previously proved results and, in a few places, select CVC5 instead of Z3 for steps where it performs better. The companion checks both top-level theorems, including their dependencies; its log records timings as diagnostics rather than benchmarks.

What to take away. The development reuses every technique from the earlier sections at larger scale. Generalization appears again: the central lemma `tautologyImpliesEval` is stated for arbitrary variable assignments, not only for the empty list of assumptions with which `isTautology` starts. New techniques are explicit termination measures and strong induction. The structure of the proof follows the structure of the program: one group of lemmas per auxiliary function, and top-level theorems assembled from them by short calculations. SBV's documentation contains about forty further TP developments, including sorting algorithms, a semantics-preserving constant folder, and a compiler from expressions to a stack machine.

14 FROM VERIFIED CODE TO C

The symbolic programs we have written can also be compiled. SBV translates a symbolic function to C code that runs without a solver. For the saturating adder of Section 3:

```
generateAdder :: FilePath -> IO ()
generateAdder dir = compileToC (Just dir) "sat_add" $ do
  cgOverwriteFiles True
  cgGenerateDriver False
  x <- cgInput "x"
  y <- cgInput "y"
  cgReturn $ satAdd x y
```

This produces `sat_add.c` and a matching header. Because the C code is generated from the same `satAdd` that we proved correct, there is no manual rewrite in which the two could drift apart. For this adder, the generated header maps `SWord8` to `uint8_t`, and the arithmetic helpers implement 8-bit modular addition and subtraction using unsigned intermediates and masks.

Code generation is not a verified compiler, and it is worth being precise about what is known. The proof establishes a property of the *symbolic* function. The C program is a separate artifact, produced by a translation that must implement each operation faithfully. For a function with a small input domain we can check the compiled artifact exhaustively: the companion compiles the generated adder with undefined-behavior sanitization and compares it on all 65,536 inputs with an independent oracle written directly in C. The proof and the exhaustive test make different claims,

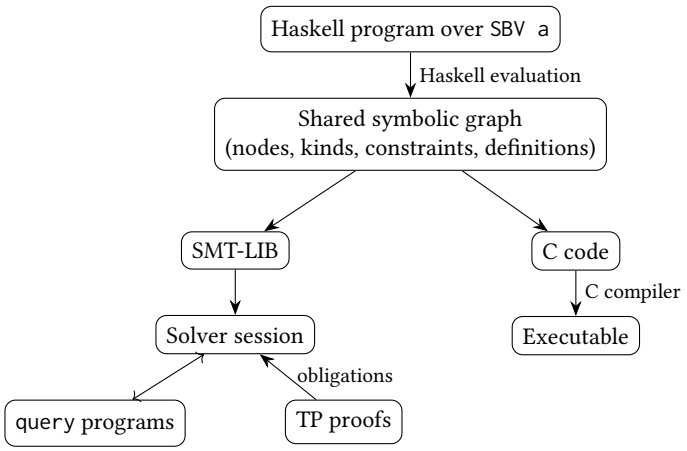


Fig. 1. Haskell evaluation builds a shared graph, which is translated to SMT-LIB for the solver or to C for execution. Queries and proofs drive the solver.

about the model and about this compiled code. Exhaustive agreement covers the tested executable and input domain; it is not a proof of the backend for all programs, compilers, or target platforms.

Not all of SBV's vocabulary is executable. Quantifiers and uninterpreted sorts, for example, have no operational meaning, and the code generator rejects programs that use them rather than approximating them. Uninterpreted functions become calls to C functions that the user provides, which then join the trusted code.

15 UNDER THE HOOD

This section explains enough of SBV's implementation to make its behavior predictable. Figure 1 shows the overall flow.

Symbolic values are deferred graph nodes. Internally, a value of type `SBV a` is either a concrete constant or a deferred computation that, given the state of the current symbolic run, adds a node to a graph and returns its identifier:

```
newtype SBV a = SBV SVal
data SVal     = SVal Kind (Either CV (Cached SV))
newtype Cached a = Cached (State -> IO a)
```

This representation lets `\x y -> x + y` have the pure type `SWord8 -> SWord8 -> SWord8` while still allocating graph nodes. Operations on constants are folded immediately, which is why `satAdd 250 10` evaluates to `255` without a solver. The run-time `Kind` records the logical sort, so simplifications can respect it: rewriting `x .== x` to true is valid for integers but not for floats.

Sharing comes in two forms. A generator that combines a term with itself, such as `iterate (\t -> t + t) x !! 30`, describes a term that is exponentially large as a tree but linear as a graph. SBV preserves this sharing in two ways. It memoizes each deferred computation, using stable names in the style of observable sharing [Gill 2009], so that forcing the same Haskell value twice yields the same node. It also interns nodes, so that separately constructed applications of the same operator to the same arguments reuse a node. Neither mechanism identifies arbitrary mathematically equal expressions.

Nodes belong to a run. A graph node is meaningful only in the run that created it. Two values of type `SInteger` from different calls to `prove` cannot be combined, even though their types agree, and SBV checks for common mistakes of this kind. To carry a result from one run to another, extract a concrete value from a model and turn it back into a literal. Section 8 illustrates the same concrete-to-symbolic conversion with `literal v`, though its loop stays within one run. Constants and reusable function definitions can be shared across runs; session-specific symbolic unknowns cannot.

Queries synchronize dependencies. When a query creates a new term, SBV records the declarations it requires: sorts for datatypes (including those nested inside lists, tuples, or other datatypes), uninterpreted constants, and definitions of `smtFunctions` it calls. Before the term is used, any of these that the solver has not seen are sent. This is why a datatype can first appear inside a query. The companion exercises this case with two fresh values of a parameterized datatype created by `freshVar`.

15.1 Termination is a proof obligation

The `Functions proven terminating` line in TP output records a check that is separate from the theorem's induction. SMT functions denote total functions. An arbitrary recursive equation need not define one: for example, $b(n) = b(n) + 1$ has no integer-valued solution. Admitting that equation as a definition would make the logical context inconsistent, allowing unrelated claims to be proved. Haskell's willingness to accept a recursive definition is therefore insufficient for using it in a proof.

When SBV registers a recursive `smtFunction`, it records a deferred termination check along with the definition. Automatic checks try measures suggested by the argument types, such as list length or datatype size. For a measure m and each recursive call from arguments $\vec{x}$ to $\vec{x}'$, the obligations have the form

$$m(\vec{x}) \geq 0, \qquad C(\vec{x}) \Longrightarrow m(\vec{x}') < m(\vec{x}),$$

where C is the condition under which that call is reached. The ordering must be well founded: nonnegative integers cannot decrease forever. For `sumL`, a recursive call is reached only in the nonempty-list branch, where the tail is one element shorter. Checking a proper datatype subterm can also be done structurally, without a solver query. Mutual recursion is checked across calls between members of the recursive group, not just self-calls.

TP gathers these checks from the definitions encountered while constructing a lemma or calculation, including functions used by its dependencies. Before checking the proof, it verifies newly encountered measures and remembers the successful checks within that TP run. At the end, `runTP` audits that no encountered check was left outstanding and reports the verified functions. This is why `sumNonNeg` reports both `sumL` and the library fold used to express its premise. These checks concern the symbolic definitions actually encountered, not every Haskell function in the source module.

If an automatic measure cannot be established, SBV rejects the definition and requests an explicit one via `smtFunctionWithMeasure`; an unknown solver answer does not establish termination. The normalizer in Section 13 illustrates a lexicographic measure and proved helper lemmas. The companion also checks that a definition with the equation $b(n) = b(n) + 1$ is rejected. Explicitly choosing `smtFunctionNoTermination` bypasses the check, but TP records that assumption in the proof's dependencies and reports the result as `Modulo` the function's termination. Such a result is conditional, not evidence that the function terminates. Productivity checks for corecursive definitions are a separate facility, outside this tutorial's finite-list setting.

15.2 Why finite lists and datatypes?

SBV's `SList a` denotes finite SMT sequences, excluding Haskell's bottom, $\perp$, in both spine and elements. Thus `[undefined]`, `1 : undefined`, and `repeat 0` are outside the model. Every symbolic-list proof has this restriction, including at all levels of nested lists. "Arbitrary length" means no finite bound is imposed; infinity is not another possible length.

Finiteness is not specific to lists. SBV supports algebraic datatypes with recursive fields, but their SMT interpretations likewise contain only finite, total constructor trees, including with mutual or nested recursion. `SExpr` from Section 9 admits finite trees of arbitrary size; the cyclic Haskell value `e = Add (Num 1) e` has no counterpart. All fields must be defined and recursive structure must terminate. Datatype proofs therefore have the same finite, total scope as list proofs. This restriction concerns recursive constructor structure. Arrays instead use the total-mapping semantics of Section 7, with potentially infinite index domains.

The distinction matters even for familiar equations. Reversing twice is the identity on finite lists, but consider the ordinary Haskell value `xs = repeat 0`. The expression `take 1 xs` produces `[0]`, whereas `take 1 (reverse (reverse xs))` diverges: the first reversal never reaches the end. A finite-list theorem therefore cannot justify this rewrite for an arbitrary lazy Haskell list. Likewise, a termination argument based on decreasing length presupposes a finite length; it does not establish productivity on infinite input.

We choose this boundary because it aligns reasoning with the solver's finite sequences and inductive datatypes, giving TP a direct structural induction principle and a natural termination measure. It covers the folds, accumulator algorithms, and transformations of finite data developed here. Extending the domain would require additional semantics and proof rules: reasoning about partial approximations must account for definedness and strictness, while infinite streams require a suitable representation and coinductive reasoning. Merely adding a bottom case to list induction cannot create infinite values in the SMT sequence sort. Keeping these concerns outside the sequence and inductive datatype models limits the machinery whose soundness these proofs depend on.

This choice is consistent with the motivation of Danielsson et al. [2006]. For their specified languages, equality proved in a total semantics yields a relation between values in a partial semantics. For their example on lists of natural numbers, this recovers equality on total, finite inputs; at function types, the relation compares behavior on related inputs. Their result explains how total reasoning can avoid repeatedly propagating definedness conditions. It is not a transfer theorem for SBV or all of Haskell: applying an SBV result to a Haskell implementation still requires matching its operations and input domain to the symbolic model.

When a problem concerns streams, a separate model may suffice. For example, representing a stream by an index function $s : \mathbb{N} \to A$ turns equality into pointwise equality and suits uniform operations such as map and zip. That is a different specification, not an extension of a theorem over `SList`; operations such as filtering require further reasoning.

16 PITFALLS AND WHAT TO TRUST

Common pitfalls.

- *Wrong domain.* A property proved over `SInteger` says nothing about `SInt32`, and vice versa. Choose the type that matches the code you care about. List and recursive-datatype theorems cover finite, total values only (Section 15.2).
- *Specification repeats the implementation.* Write specifications in a different style or a wider domain (Section 3).

- *Haskell* `if` *or recursion on symbolic data.* Use `ite` and `sCase` for symbolic choices, and `smtFunction` for recursion over symbolic structure (Section 6).
- *Contradictory assumptions.* If `constrained` assumptions are unsatisfiable, everything is provable. A quick `sat` on the assumptions alone detects this.
- *Treating unknown as an answer.* A timeout or unknown result is neither a proof nor a counterexample.
- *Too specific an induction statement.* If a proof step involving the induction hypothesis fails, the statement probably needs generalizing (Sections 10 and 12).
- *Mixing runs.* Unknowns belong to the run that created them; move concrete values instead (Section 15).

What a proof rests on. A Q.E.D. from SBV depends on the correctness of SBV's translation to SMT-LIB, the solver, and, for TP, the generation of proof obligations. TP proofs also depend on any axioms the user introduces and on any functions whose termination checks were explicitly bypassed. SBV tracks these dependencies: a proof that rests on an axiom, or on an unfinished step marked with `sorry`, is reported as `Modulo` the offending lemmas rather than as `Proven`. TP does not replay solver certificates in an independent proof kernel, so its guarantees are those of the underlying solver. For generated C, the C compiler and the code generator join the trusted base, which is why Section 14 validates the generated code separately.

17 RELATED TOOLS AND FURTHER READING

QuickCheck [Claessen and Hughes 2000] and SBV ask similar questions with different evidence: a passing test campaign samples the input space, while an unsatisfiability result covers it. The two combine well, and SBV's own test suite compares concrete evaluation with solver results using QuickCheck.

Several systems embed solver-aided programming in a host language. Rosette [Torlak and Bodík 2014] provides a symbolic virtual machine for Racket that lifts ordinary host constructs to symbolic values. Grisette [Lu and Bodík 2023] is a Haskell library for symbolic compilation that focuses on efficient symbolic evaluation of programs with state merging. Smten [Uhler and Dave 2013] translated high-level Haskell-like symbolic computations into SMT queries. The what4 library [Galois, Inc. 2026] offers a typed Haskell interface to many solvers and is used inside verification tools. SBV makes the boundary between Haskell and symbolic control explicit through its types and operators, and adds the TP proof layer and C generation on the same representation.

For proofs about Haskell programs, Liquid Haskell [Vazou et al. 2014] checks refinement types with an SMT solver, and refinement reflection [Vazou et al. 2018] extends it to equational proofs about user-defined functions. There, specifications are refinement types on the program itself and proofs are Haskell terms checked against them; SBV's TP proofs are instead separate Haskell programs about symbolic models. Dafny [Leino 2010] shows how far the combination of user-supplied invariants and automatic verification-condition discharge can go in a dedicated language.

The SBV distribution contains a large collection of documented examples under `Documentation.SBV.Examples`, covering puzzles, cryptography, code generation, optimization, and theorem proving. The TP examples mentioned in Section 13 are a good next step after this tutorial.

18 CONCLUSION

SBV connects constraint solving, optimization, and inductive proofs through one vocabulary of symbolic values. Effective use depends on choosing the right semantic domain, writing independent specifications, separating Haskell from symbolic control, and generalizing induction hypotheses. The companion provides executable models and checked calculations that readers can adapt to their own programs.

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